Developer Guide

This section is for people working on the SWAT+ Fortran source: building from a clone, running the test suite, and submitting changes back to `swat-model/swatplus`.

If you only need to run a prebuilt SWAT+ executable, see Install instead.

Pages

- Build with CMake. The build used on every platform: configuration, compilers (gfortran, `ifx`, `ifort`), compiler options, debug and release builds.
- Visual Studio on Windows. Visual Studio 2022 with the Intel oneAPI Fortran compiler: cloning, configuring, and debugging against an input dataset.
- VS Code Codespaces. Running a SWAT+ build and debug session inside a GitHub Codespaces virtual instance. Useful when you do not want to install a Fortran toolchain locally.
- Testing. The scenario regression test that compares model output with reference outputs, and how to run it at 62.0.1.
- Tagging and releases. How the version number comes from the Git tag and reaches the executable, and how a release is made.
- Coding conventions. Fortran 90/95 style rules for SWAT+ source contributions: portability, readability, reliability, arrays, I/O, and obsolescent features to avoid.
- Contributing. Reporting problems, pull requests, and contributing to this documentation.
- Code of conduct. The Contributor Covenant, applied to the SWAT+ project community.